

每日一題50

單元4 數列與級數-等比數列

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114翰林第二次模考※4

設 a, b, c 均為正整數，且 $1 < a < b < c < 9$ ，若 $\frac{a}{5}, \frac{b}{10}, \frac{c}{15}$ 三數成等比，則公比為何？

(1) 2

☒ (2) 1

(3) 0.5

(4) 0.25

(5) 0.125

<Sol>

$$\because \frac{a}{5}, \frac{b}{10}, \frac{c}{15} \text{ 成等比}$$

$$\therefore \left(\frac{b}{10}\right)^2 = \frac{a}{5} \cdot \frac{c}{15} \Rightarrow \frac{b^2}{100} = \frac{ac}{75} \Rightarrow 75b^2 = 100ac$$

$$\Rightarrow 3b^2 = 4ac$$

$$\left[\begin{array}{l} \because 3b^2 = 4ac = 2(2ac) \Rightarrow 3b^2 \text{ 必偶數} \\ \Rightarrow b^2 \text{ 必偶數} \Rightarrow b = 4, 6 \end{array} \right]$$

$$\textcircled{1} b=4, 3 \cdot 4^2 = 4ac \Rightarrow ac=12$$

$$\Rightarrow a=2, c=6 \text{ (合)}$$

$$\textcircled{2} b=6, 3 \cdot 6^2 = 4ac \Rightarrow ac=27$$

$$\Rightarrow a=3, c=9 \text{ (不合)}$$

$$\therefore a=2, b=4, c=6$$

$$\Rightarrow r = \frac{\frac{6}{10}}{\frac{2}{5}} = \frac{\frac{4}{10}}{\frac{2}{5}} = 1 \quad \#$$